Chapter 5. Association and Independence of Two Variables

Pearson and Spearman correlation, Kendall's tau, permutation tests of independence

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5.1 Pearson correlation

For a pair (X, Y), $\rho = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y} \in [-1,1]$ measures *linear* association. Independence implies $\rho = 0$; the converse holds under bivariate normality but not in general (e.g. $Y = X^2$ with X symmetric gives $\rho = 0$). From pairs (X_i, Y_i), $i = 1, \dots, n$, the sample correlation is $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$. The least squares slope is $\hat\beta = S_{XY}/S_{XX} = r\sqrt{S_{YY}/S_{XX}}$. For the population, $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}\,X = \rho\,\sigma_Y/\sigma_X$, so testing $\beta = 0$ in a simple regression and testing $\rho = 0$ are the same problem.

Proposition 5.1

If (X_i, Y_i) are i.i.d. bivariate normal with $\rho = 0$, then $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$.

Proof. Condition on the X's. Then Y_i are i.i.d. normal, i.e. the regression model with $\beta = 0$, and the slope t-statistic $\hat\beta/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$. With $SSE = S_{YY}(1-r^2)$ and $\widehat{\mathrm{se}}^2 = SSE/((n-2)S_{XX})$, one gets $\hat\beta/\widehat{\mathrm{se}} = r\sqrt{(n-2)/(1-r^2)}$. The conditional law does not depend on the X's. ∎

Permutation test of independence. Under H_0: $X \perp Y$, the Y's can be permuted among the fixed X's without changing the joint law, so shuffling Y gives an exact test with any statistic, e.g. r. The alternative it is sensitive to is linear association. Note that H_0 is independence, not $\rho = 0$: if $\rho = 0$ but X, Y are dependent, shuffling is not justified.

Proposition 5.2

Over all $n!$ permutations of $(Y_1, \dots, Y_n)$ with the X's fixed, the permutation distribution of r has mean 0 and variance $1/(n-1)$, whatever the data.

Proof. Let $a_i = X_i - \bar X$, $b_i = Y_i - \bar Y$ and π uniform. Then $r = \sum_i a_i b_{\pi(i)}/\sqrt{S_{XX}S_{YY}}$ and $Eb_{\pi(i)} = 0$. Also $\mathrm{Var}\,b_{\pi(i)} = S_{YY}/n$ and $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = -S_{YY}/(n(n-1))$ for $i \neq j$. With $\sum_{i\neq j} a_i a_j = -S_{XX}$,

$$\mathrm{Var}\left(\sum a_i b_{\pi(i)}\right) = \frac{S_{XX}S_{YY}}{n} + \frac{S_{XX}S_{YY}}{n(n-1)} = \frac{S_{XX}S_{YY}}{n-1}.$$

∎

Combined with a finite-population CLT, $\sqrt{n-1}\,r$ is approximately $N(0,1)$ under the permutation distribution, a distribution-free counterpart of Proposition 5.1.

5.2 Spearman's rank correlation

Replace X_i, Y_i by their ranks R_i, S_i and compute the Pearson correlation: $r_s = \mathrm{Corr}(R, S)$. It measures *monotone* association, is invariant under increasing transformations of either variable, and is robust to outliers.

Proposition 5.3

Without ties, $r_s = 1 - \dfrac{6\sum_i (R_i - S_i)^2}{n(n^2-1)}$.

Proof. Both rank vectors are permutations of $1, \dots, n$, with mean $\frac{n+1}{2}$ and $\sum(R_i - \bar R)^2 = \frac{n(n^2-1)}{12}$. Since $\sum(R_i - S_i)^2 = \sum(R_i - \bar R)^2 + \sum(S_i - \bar S)^2 - 2\sum(R_i - \bar R)(S_i - \bar S)$, we get $\sum(R_i - \bar R)(S_i - \bar S) = \frac{n(n^2-1)}{12} - \frac{1}{2}\sum(R_i - S_i)^2$. Divide by $\frac{n(n^2-1)}{12}$. ∎

The permutation test with r_s tests independence against monotone association. It is distribution-free, and Proposition 5.2 applies to it verbatim.

5.3 Kendall's tau

Motivation. Pearson's correlation measures linear association and is sensitive to outliers. Kendall's tau counts concordant minus discordant pairs, so it measures monotone association and depends only on ranks.

A pair of observations $i < j$ is **concordant** if $(X_i - X_j)(Y_i - Y_j) > 0$ and **discordant** if < 0. Let $U_{ij} = \operatorname{sgn}[(X_i - X_j)(Y_i - Y_j)]$.

Definition (Kendall's tau)

Sample: $\tau_n = \binom{n}{2}^{-1} \sum_{i<j} U_{ij} = (N_c - N_d)/\binom{n}{2}$. Population: for independent copies $(X, Y), (X', Y')$, $\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\}$.

τ_n is unbiased for τ. Unlike r_s, it has a clear population counterpart.

Example 5.1

$x = (1, 2, 3, 4, 5)$, $y = (2, 1, 4, 3, 5)$. Only the pairs $(1, 2)$ and $(3, 4)$ are discordant, so $\tau_n = (8 - 2)/10 = 0.6$. The rank differences are $(-1, 1, -1, 1, 0)$, so by Proposition 5.3, $r_s = 1 - 6 \cdot 4/120 = 0.8$. Pearson's r is also 0.8 here, since the data are their own ranks.

Proposition 5.4 (Normal case)

If (X, Y) is bivariate normal with correlation ρ, then $\tau = \frac{2}{\pi} \arcsin \rho$.

Proof [Supplement]. $(U, V) = (X - X', Y - Y')$ is centered normal with correlation ρ. Write $U = Z_1$ and $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$, then pass to polar coordinates $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$ with Θ uniform, and set $\rho = \sin\phi$. Then $V = R\sin(\Theta + \phi)$. The set where $\cos\Theta$ and $\sin(\Theta + \phi)$ share a sign has total angle $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\phi}{\pi}$ and $\tau = 2P(UV > 0) - 1 = \frac{2\phi}{\pi}$. ∎

This gives a robust estimator $\hat{\rho} = \sin(\pi\tau_n/2)$ of a normal correlation in the presence of outliers.

Proposition 5.5

If $X \perp Y$ with continuous margins, then $E\tau_n = 0$ and $\operatorname{Var}\tau_n = \dfrac{2(2n + 5)}{9n(n - 1)}$. Moreover $\tau_n / \sqrt{\operatorname{Var}\tau_n} \xrightarrow{d} N(0, 1)$.

Proof [Supplement]. Each $U_{ij} = \pm 1$ with probability $\frac{1}{2}$, so $\operatorname{Var} U_{ij} = 1$. Pairs with no common index give independent U's. For a shared index, independence of X and Y gives $E(U_{12}U_{13}) = \left(E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\right)^2 = \left(E(2V - 1)^2\right)^2 = \frac{1}{9}$ with V uniform. There are $\binom{n}{2} \cdot 2(n - 2)$ ordered pairs of pairs sharing one index, so

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2}\Big(1 + \frac{2(n-2)}{9}\Big) = \binom{n}{2}\frac{2n+5}{9},$$

and dividing by $\binom{n}{2}^2$ gives the formula. Asymptotic normality follows from Hoeffding's theory of U-statistics. ∎

Under independence the exact null distribution of τ_n is its permutation distribution (shuffle Y), which is distribution-free. Without ties this is Kendall's tabulated null distribution. So there are three equivalent ways to test independence:

(i) a (random) permutation test;

(ii) Kendall's exact null distribution;

(iii) the normal approximation of Proposition 5.5, already good for $n \approx 10$ except in the extreme tails. Pairs with $(X_i - X_j)(Y_i - Y_j) = 0$ give $U_{ij} = 0$. Software usually reports the tie-corrected version $\tau_b = (N_c - N_d)/\sqrt{(N_0 - T_X)(N_0 - T_Y)}$, where $N_0 = \binom{n}{2}$ and T_X, T_Y count the pairs tied in X or in Y. With ties only the permutation test remains exact.

Which measure? Under bivariate normality, use the t-test of r. With heavy-tailed margins but linear dependence, rank-based r_s and τ_n give more powerful tests. For monotone nonlinear dependence, r_s or τ_n are the natural measures. All three can be near zero for strong non-monotone dependence (e.g. $Y = \sin(3X) + \epsilon$), so failing to reject independence does not mean independence.

5.4 Selected exercises

Motivation. The exercises justify shuffling in regression, give a dependent pair with $\rho = \tau = 0$, and derive the null distribution of Kendall's statistic from a decomposition of permutations into independent uniform pieces.

Exercise 5.2 (When shuffling is justified)

Let $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ with (X_i, ϵ_i) i.i.d., X_i independent of ϵ_i, $0 < \operatorname{Var} X_i < \infty$ and $\operatorname{Var} \epsilon_i < \infty$. Show that (X_1, Y_1) and (X_1, Y_2) have the same joint distribution if $\beta_1 = 0$, and different joint distributions if $\beta_1 \neq 0$.

Solution.

If $\beta_1 = 0$: $(X_1, Y_1) = (X_1, \beta_0 + \epsilon_1)$ with X_1 independent of ϵ_1, so its law is the product of the law of X and the law of $\beta_0 + \epsilon$. Also $(X_1, Y_2) = (X_1, \beta_0 + \epsilon_2)$ with X_1 independent of ϵ_2 (different units), so it has the same product law.

If $\beta_1 \neq 0$: $\operatorname{Cov}(X_1, Y_1) = \beta_1 \operatorname{Var} X_1 + \operatorname{Cov}(X_1, \epsilon_1) = \beta_1 \operatorname{Var} X_1 \neq 0$, while $\operatorname{Cov}(X_1, Y_2) = 0$ because X_1 is independent of (X_2, ϵ_2). Covariances are determined by the joint law (the moments are finite), so the two laws differ.

Shuffling the Y's among the fixed X's replaces pairs (X_i, Y_i) by pairs $(X_i, Y_{\pi(i)})$. This preserves the joint law of the data exactly when $\beta_1 = 0$, which is the null hypothesis of the permutation test. ■

Exercise 5.5 (Dependence with $\rho = \tau = 0$)

Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$. Show that X and Y are dependent but $\rho = 0$ and $\tau = 0$.

Solution [Supplement].

Dependence. $Y < \frac{1}{4}$ iff $|X| < \frac{1}{2}$, so $P(X > \frac{1}{2}, Y < \frac{1}{4}) = 0$, while $P(X > \frac{1}{2})P(Y < \frac{1}{4}) = \frac{1}{4} \cdot \frac{1}{2} \neq 0$.

Pearson. X is symmetric about 0 with bounded moments, so $EX = EX^3 = 0$ and $\operatorname{Cov}(X, Y) = EX^3 - EX\, EX^2 = 0$.

Kendall. For an independent copy (X', Y'),

$$(X - X')(Y - Y') = (X - X')(X^2 - X'^2) = (X - X')^2(X + X').$$

Since $P(X = X') = 0$, its sign is a.s. the sign of $X + X'$, a continuous variable symmetric about 0. So both signs have probability $\frac{1}{2}$, and $\tau = 0$.

Since τ_n is unbiased for τ, $E\tau_n = 0$ here as well, so τ_n is centered exactly as under independence. Tests based on r, r_s or τ_n are aimed at monotone association and have little power against such a U-shaped dependence. ■

Exercise 5.6 (Kendall's null distribution via inversions)

Assume no ties. List the pairs in increasing order of X and let $\pi(k)$ be the rank of the Y-value of the kth pair, so π is a permutation of $\{1, \dots, n\}$. Let N_d be the number of discordant pairs.
(a) Show that $N_d = \#\{j < k : \pi(j) > \pi(k)\}$ and $\tau_n = 1 - 4N_d/(n(n-1))$.
(b) Let $I_k = \#\{j < k : \pi(j) > \pi(k)\}$. Show that if π is uniform on the $n!$ permutations, then $I_1, \dots, I_n$ are independent and I_k is uniform on $\{0, \dots, k-1\}$.
(c) Deduce $E\tau_n = 0$ and $\operatorname{Var} \tau_n = 2(2n+5)/(9n(n-1))$ under independence of X and Y.

Solution [Supplement].

(a) For $j < k$ in the X-order, X increases from pair j to pair k, so the pair is discordant iff Y decreases, i.e. $\pi(j) > \pi(k)$. Without ties $N_c + N_d = \binom{n}{2}$, so $\tau_n = (\binom{n}{2} - 2N_d)/\binom{n}{2} = 1 - 4N_d/(n(n-1))$.

(b)

Step 1. The map $\pi \mapsto (I_1, \dots, I_n)$ goes from the $n!$ permutations into $\prod_{k=1}^{n} \{0, \dots, k-1\}$, which also has $n!$ elements. It suffices to show that it is injective.

Step 2. We recover π from the I's backwards. Suppose $\pi(k+1), \dots, \pi(n)$ are known. Then the set $V_k = \{\pi(1), \dots, \pi(k)\}$ of the remaining values is known. Exactly I_k elements of V_k exceed $\pi(k)$, so $\pi(k)$ is the $(k - I_k)$th smallest element of V_k. Starting from $k = n$ (where $V_n = \{1, \dots, n\}$), this determines π completely.

Step 3. So the map is a bijection. A uniform π is therefore mapped to a uniform point of the product set, which means the coordinates I_k are independent and I_k is uniform on $\{0, \dots, k-1\}$.

(c)

Step 1. Under independence with continuous margins, the Y's attached to the X-ordered pairs are i.i.d. continuous given the X's, so their rank vector π is uniform.

Step 2. $N_d = \sum_k I_k$ with $EI_k = \frac{k-1}{2}$ and $\operatorname{Var} I_k = \frac{k^2-1}{12}$. So $EN_d = \frac{n(n-1)}{4}$ and $E\tau_n = 0$. Also

$$\operatorname{Var} N_d = \frac{1}{12} \sum_{k=1}^{n} (k^2 - 1) = \frac{1}{12} \Big(\frac{n(n+1)(2n+1)}{6} - n \Big) = \frac{n(n-1)(2n+5)}{72}.$$

Step 3. Hence $\operatorname{Var} \tau_n = \frac{16}{n^2(n-1)^2} \cdot \frac{n(n-1)(2n+5)}{72} = \frac{2(2n+5)}{9n(n-1)}$, as in Proposition 5.5.

The decomposition also gives the exact null law through the generating function $Eq^{N_d} = \prod_{k=1}^{n} \frac{1+q+\cdots+q^{k-1}}{k}$. It gives an elementary proof of asymptotic normality as well: the I_k are independent with $\sum_k E|I_k - EI_k|^3 \le \sum_k k^3 = O(n^4)$, while $(\operatorname{Var} N_d)^{3/2} \asymp n^{9/2}$, so Lyapunov's condition holds. ■